Nonconvergence of randomized subtraction games with odd moves

Research proof and formalization packet

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Abstract

Let a finite nonempty subtraction set consist entirely of positive odd integers, and choose uniformly among the legal moves at each heap. We prove that the probability of winning for the player to move does not converge as the heap grows. This establishes the general all-odd nonconvergence assertion left open by Capitelli and Somma. An alternating inequality for consecutive probabilities gives positive initial data for a conserved finite-window sum. Convergence would force this sum to vanish. The proof requires neither coprime moves nor simplicity of characteristic roots. The complete recurrence, actual recursively defined probability sequence, positive invariant, and nonconvergence theorem are verified in Lean.

1 The game and the source question

Fix a finite nonempty set S of positive integers. From a heap of n chips, a legal move subtracts an element of $E_n = \{k \in S : k \le n\}$. The next move is chosen uniformly from E_n; a player with no legal move loses. Let a_n denote the probability that the player to move wins. Thus

$$a_n = \begin{cases} 0, & E_n = \varnothing, \\ 1 - \dfrac{1}{|E_n|} \displaystyle\sum_{k \in E_n} a_{n-k}, & E_n \neq \varnothing. \end{cases} \tag{1}$$

Positive moves make this a uniquely determined recursion; in particular $a_0 = 0$ and $0 \le a_n \le 1$ by induction.

Capitelli and Somma [1] analyze this exact recurrence. They settle sets of cardinality below three and prove convergence in their coprime setting when an even move is present. At the start of Section 3 they explicitly leave general all-odd nonconvergence open; Question 3.4 asks about the nonzero coefficient of the root -1 in that setting. We give a direct proof of the missing nonconvergence assertion.

Theorem 1. *For every finite nonempty set S of positive odd integers, the sequence (a_n) defined by (1) has no real limit.*

This is one universal source-question resolution. The singleton and two-move cases are included for a complete statement and receive no separate novelty claim. The neighboring Conjecture 3.3, indexed as MathDB 363319, concerns simplicity of all characteristic roots. The present theorem does not assert or require that conjecture.

2 Consecutive-probability inequalities

Lemma 2. *If every move in S is odd, then for every $m \geq 0$,*

$$a_{2m} \leq a_{2m+1}, \qquad a_{2m+2} \leq a_{2m+1}.$$

Proof. Prove the inequalities successively by induction. The first begins with $a_0 = 0 \leq a_1$. At heaps $2m+1$ and $2m+2$ the legal sets coincide, because no even move can become newly legal. Each odd subtraction gives an earlier even heap and its following odd heap. The already proved first inequalities make the latter follower probability at least the former. Subtracting equal-sized averages in (1) gives $a_{2m+2} \leq a_{2m+1}$. If the legal set is empty, both probabilities are zero.

Next compare heaps $2m+2$ and $2m+3$. For every old legal move the two followers are an earlier odd heap and its following even heap. Their probabilities satisfy the just-established second inequalities, so the new follower sum is no larger. The only possible newly legal move is $2m+3$ itself; its follower has probability $a_0 = 0$. Adding this zero cannot increase the average. Hence $a_{2m+2} \leq a_{2m+3}$. If the old legal set is empty, its probability is zero and the new probability is nonnegative. This covers the initial empty-set branch and completes the induction. □

Put

$$c_n = (-1)^n(1 - 2a_n), \qquad C(h) = \sum_{j=0}^{h-1} c_j, \qquad C(0) = 0.$$

Since every $k \in E_n$ is odd, multiplying the balance form of (1) by $(-1)^n$ gives

$$|E_n|c_n = \sum_{k \in E_n} c_{n-k} \qquad (n \geq 0). \tag{2}$$

It holds also for empty legal sets as $0 = 0$. For odd $h = 2q+1$, Lemma 2 gives

$$C(h) = 1 + 2\sum_{j=1}^{q}(a_{2j-1} - a_{2j}) \geq 1. \tag{3}$$

3 A positive conserved sum

Let $D = \max S$ and $t = |S|$. Define, for $n \geq 0$,

$$I_n = \sum_{k \in S}\big(C(n+D) - C(n+D-k)\big) = \sum_{k \in S}\sum_{j=0}^{k-1} c_{n+D-k+j}. \tag{4}$$

Lemma 3. *For every $n \geq 0$,*

$$I_n = I_0 = \sum_{k \in S} C(k) \geq t > 0.$$

Proof. Every move is legal at heap $n+D$. Subtracting consecutive windows and using (2) gives

$$I_{n+1} - I_n = tc_{n+D} - \sum_{k \in S} c_{n+D-k} = 0.$$

It remains to compute the initial value. Sum (2) over $0 \leq n < D$ and exchange finite sums. The left-hand side becomes

$$\sum_{k \in S}\sum_{n=k}^{D-1} c_n = \sum_{k \in S}\big(C(D) - C(k)\big),$$

whereas the right-hand side is

$$\sum_{k\in S}\sum_{n=k}^{D-1} c_{n-k} = \sum_{k\in S} C(D-k).$$

Rearranging gives

$$I_0 = \sum_{k\in S}\big(C(D) - C(D-k)\big) = \sum_{k\in S} C(k).$$

The equality is for the total sums; individual suffix and prefix windows need not agree. Since every $k \in S$ is odd, (3) yields $I_0 \geq |S| > 0$. □

Proof of Theorem 1. Suppose $a_n \to L$. For $n \geq D$, (1) uses the full set S. Taking limits in its finite average gives $L = 1 - L$, hence $L = 1/2$. Consequently $|c_n| = |1 - 2a_n| \to 0$. Each of the finitely many shifted terms in the second expression (4) then tends to zero, so $I_n \to 0$. This contradicts Lemma 3. □

4 Formal scope, prior work, and novelty limits

The complete Lean endpoint is `OddSubtraction.all_odd_game_nonconvergence`. It quantifies over an arbitrary finite set, requires nonemptiness and odd positive moves, and proves that there is no real limit for the actual recursive `gameWin` sequence. No supplied probability model, invariant, spectral result, gcd condition, or root-simplicity hypothesis remains. The second main endpoint proves the quantitative invariant bound $I_n \geq |S|$. The packet contains six modules and a pinned isolated replay of all 37 explicitly inventoried declarations, including the model structure. The only reported axioms are `propext`, `Classical.choice`, and `Quot.sound`.

The earlier noisy-channel game framework [2] supplies related recurrences and fairness questions, but not this all-odd nonconvergence theorem. Work on ultimately bipartite subtraction games [3] concerns optimal-play nim values and related parity phenomena; the present probabilities average all legal moves rather than select optimal strategies. The elementary finite-window conservation argument is a tool; no separate discovery is claimed for it or the known small cases.

Independent review checked the exact primary game definition, all initial branches, finite-sum indices, nonconvergence argument and final formal endpoint. Searches of the source title, identifier, author pair and related randomized-subtraction literature found no earlier full resolution. The current source record has one version and no journal reference. These are documented checks, not an absolute certification of priority; unindexed or unpublished work may remain.

Our formal certificate proves nonconvergence directly. It does not formalize characteristic-root asymptotics or assert that all roots are simple. In particular, it must not be described as a proof of the different MathDB 363319 statement. The source's proposed spectral route is bypassed by the positive invariant.

References

[1] N. Capitelli and F. Somma, *On sequences arising from randomizing subtraction games*, arXiv:2405.19593v1 (2024), Sections 1, 3, and 4.1. Primary source.

[2] N. A. Capitelli and M. L. Privitelli, *Playing through a noisy channel (and knowing it)*, arXiv:2012.06768v1 (2020). Primary source.

[3] G. Cairns and N. B. Ho, *Ultimately bipartite subtraction games*, Australasian J. Combinatorics 48 (2010), 213–220. Primary source.